

IoT operating systems include:

Real-time capabilities.

- *Time-sensitive task management.* IoT devices often operate in time-critical environments, requiring rapid responses to events. Real-time operating systems (RTOS) are designed to prioritise and execute tasks based on their urgency, ensuring timely completion. This is crucial for applications like industrial automation, where precise timing is essential to prevent accidents and optimise processes.

- *Deterministic behaviour.* RTOS provides predictable response times, enabling developers to design systems with reliable and consistent performance. This is particularly important for applications such as autonomous vehicles, where timely responses to sensor inputs are critical for safety.

Energy efficiency.

- *Power management.* IoT devices, especially battery-powered ones, must be energy-efficient to maximise their operational life. IoT OSes incorporate advanced power management techniques, such as dynamic voltage and frequency scaling, to optimise power consumption based on the device's workload.

- *Low-power modes.* These OSes support various low-power modes, including sleep, deep sleep, and hibernation. By transitioning to these modes when the device is idle or inactive, energy consumption can be significantly reduced, extending battery life.

Scalability.

- *Diverse hardware support.* IoT devices come in a wide range of form factors and processing capabilities. Scalable IoT OSes can adapt to these diverse hardware configurations, from low-power microcontrollers to more powerful processors.

- *Modular architecture.* A

modular design allows for easy customisation and extension of the OS to meet specific application requirements. This flexibility enables developers to tailor the OS to the needs of their IoT solution, optimising performance and resource utilisation.

Security.

- *Robust security features.* IoT devices are often vulnerable to cyberattacks due to their increasing connectivity and potential access to sensitive data. IoT OSes must implement robust security measures, such as secure boot processes, encryption, and secure communication protocols, to protect against unauthorised access and data breaches.

- *Regular security updates.*

Regular security updates are essential to address vulnerabilities and keep the OS secure. These updates should be delivered efficiently and without disrupting the device's operation.

Interoperability.

- *Protocol support.* IoT devices need to communicate with a variety of networks and protocols, including Wi-Fi, Bluetooth, Zigbee, and cellular networks. IoT OSes support these protocols to enable seamless connectivity and data exchange.

- *Cloud integration.* Seamless integration with cloud platforms allows for remote management, data analysis, and machine learning. This enables IoT devices to leverage the power of cloud computing to provide advanced features and insights.

Traditional operating systems, such as Windows, macOS, and Linux, are designed for general-purpose computing. They often have high resource requirements and may not be suitable for resource-constrained devices. IoT operating systems, on the other hand, are optimised for specific use cases and

are designed to operate efficiently on low-power hardware.

Importance in resource-constrained environments

IoT devices often operate in resource-constrained environments, where power, memory, and processing power are limited. IoT operating systems are specifically designed to address these constraints. They employ techniques such as:

- *Power management.* To conserve battery life, IoT OSes utilise low-power modes, and dynamic voltage and frequency scaling.

- *Memory optimisation.* They optimise memory usage to reduce the overall memory footprint of the system.

- *Efficient task scheduling.* IoT OSes prioritise tasks based on their importance and time constraints, ensuring that critical tasks are executed promptly.

- *Secure boot.* To protect against malicious attacks, IoT OSes often implement secure boot processes to verify the integrity of the system software.

Popular IoT operating systems

Several popular IoT operating systems have emerged to address the diverse needs of the IoT market.

- *TinyOS.* TinyOS is a lightweight and efficient operating system specifically designed for resource-constrained devices. It is ideal for sensor networks and wireless communication applications, prioritising energy efficiency and low-latency communication. TinyOS is known for its small footprint and ability to operate on devices with minimal hardware resources.

- *Contiki.* Contiki is an open source operating system with a strong focus on networking capabilities, particularly IPv6 support. It is well-suited for IoT devices that